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1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity mca_4 is
5      port(x:          in std_logic_vector(7 downto 0);
6           clk0, clk1: in std_logic;
7           r:          out std_logic_vector(3 downto 0);
8           du:         out std_logic_vector(6 downto 0));
9  end;
10
11 architecture synth of mca_4 is
12     signal reg4En, reg3En, reg2En, reg1En, reg0En, sixteen_to_fourEncoderEnable ,
13     sixteen_to_eightEncoderEnable ,
14     four_to_sixteenEncoderEnable , eight_to_fourEncoderEnable : std_logic;
15     signal eight_to_fourEncoderSignal , reg_in, output2_4De0,
16     aluOutput, rout4, rout3, rout2, rout1, rout0, out16_4En, out16_8En1,
17     out16_8En0: std_logic_vector(3 downto 0);
18     signal out4_16_de: std_logic_vector(15 downto 0);
19     signal mca4Out: std_logic_vector(6 downto 0);
20     component reg port(rin:      in std_logic_vector(3 downto 0);
21                        en, clk:   in std_logic;
22                        rout:      out std_logic_vector(3 downto 0));
23
24     end component;
25
26     component eight_to_fourEncoder port(s:          in std_logic_vector(1 downto 0);
27                                        input1, input0: in std_logic_vector(3 downto 0);
28                                        en:            in std_logic;
29                                        output8_4_en:  out std_logic_vector(3 downto 0));
30
31     end component;
32
33     component four_to_sixteenDecoder port(input4_16_de: in std_logic_vector(3 downto 0);
34                                           output4_16_de: out std_logic_vector(15 downto
35                                           0));
36
37     end component;
38
39     component alu port(reg1, reg0:          in std_logic_vector(3 downto 0);
40                       enA, enSL, enSR, enM, enN: in std_logic;
41                       aluOut:               out std_logic_vector(3 downto 0));
42
43     end component;
44
45     component two_to_fourDecoder port(input2_4De:  in std_logic_vector(1 downto 0);
46                                       en:          in std_logic;
47                                       output2_4De: out std_logic_vector(3 downto 0));
48
49     end component;
50
51     component sixteen_to_fourEncoder port(input3, input2, input1, input0: in
52     std_logic_vector(3 downto 0);
53
54     s:          in
55     std_logic_vector(1 downto 0);
56
57     en:          in std_logic;
58     output16_4En: out
59     std_logic_vector(3 downto 0));
60
61     end component;
62
63     component sixteen_to_eightEncoder port(input3, input2, input1, input0: in
64     std_logic_vector(3 downto 0);
65
66     s:          in
67     std_logic_vector(3 downto 0);
68
69     en:          in std_logic;
70     output16_8En1, output16_8En0: out
71     std_logic_vector(3 downto 0));
72
73     end component;
74
75     component display_unit port( r_in:      in std_logic_vector(3 downto 0);
76                                segments:  out std_logic_vector(6 downto 0));
77
78     end component;

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59
60 begin
61     eight_to_fourEncodersSignal (1) <= x(7) xor x(6);
62     eight_to_fourEncodersSignal (0) <= x(7) nor x(6);
63     eight_to_fourEncoderEnable <= eight_to_fourEncodersSignal (1) or
eight_to_fourEncodersSignal (0);
64
65     four_sixteenDecoder: four_to_sixteenDecoder port map(x(7 downto 4), out4_16_de(15
downto 0));
66
67     reg4En <= out4_16_de(4) or out4_16_de(5) or out4_16_de(6) or out4_16_de(8) or
out4_16_de(11);
68     reg3En <= out4_16_de(3) or output2_4De0(3);
69     reg2En <= out4_16_de(2) or output2_4De0(2);
70     reg1En <= out4_16_de(1) or output2_4De0(1);
71     reg0En <= out4_16_de(0) or output2_4De0(0);
72
73     sixteen_to_fourEncoderEnable <= x(7) and x(6);
74     sixteen_to_eightEncoderEnable <= x(7) xor x(6);
75     four_to_sixteenEncoderEnable <= x(7) xor x(6);
76
77     eight_to_fourEncoder0: eight_to_fourEncoder port map(
78         eight_to_fourEncodersSignal (1 downto 0), rout4(3 downto 0), x(3 downto 0),
79         eight_to_fourEncoderEnable , reg_in(3 downto 0));
80
81     two_to_fourDecoder0: two_to_fourDecoder port map(
82         x(1 downto 0), four_to_sixteenEncoderEnable , output2_4De0(3 downto 0));
83
84     reg4: reg port map(aluOutput(3 downto 0), reg4En, clk0, rout4(3 downto 0));
85     reg3: reg port map(reg_in(3 downto 0), reg3En, clk1, rout3(3 downto 0));
86     reg2: reg port map(reg_in(3 downto 0), reg2En, clk1, rout2(3 downto 0));
87     reg1: reg port map(reg_in(3 downto 0), reg1En, clk1, rout1(3 downto 0));
88     reg0: reg port map(reg_in(3 downto 0), reg0En, clk1, rout0(3 downto 0));
89
90     sixteen_fourEncoder: sixteen_to_fourEncoder port map(
91         rout3(3 downto 0), rout2(3 downto 0), rout1(3 downto 0),
92         rout0(3 downto 0), x(5 downto 4), sixteen_to_fourEncoderEnable , out16_4En(3 downto
0));
93
94     sixteen_eightEncoder: sixteen_to_eightEncoder port map(
95         rout3(3 downto 0), rout2(3 downto 0), rout1(3 downto 0),
96         rout0(3 downto 0), x(3 downto 0), sixteen_to_eightEncoderEnable ,
97         out16_8En1(3 downto 0), out16_8En0(3 downto 0));
98
99     logic_unit: alu port map(out16_8En1(3 downto 0), out16_8En0(3 downto 0),
100         out4_16_de(4), out4_16_de(5), out4_16_de(6), out4_16_de(8), out4_16_de(11),
101         aluOutput(3 downto 0));
102
103     r <= out16_4En(3 downto 0);
104
105     disp_unit: display_unit port map(out16_4En(3 downto 0), mca4Out(6 downto 0));
106
107     du <= mca4Out;
108
109 end;

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